

Amino Acid Conservation Scores

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The layers for assigning grades are as follows.

from -1.324 to -1.030 the grade is 9
 from -1.030 to -0.736 the grade is 8
 from -0.736 to -0.441 the grade is 7
 from -0.441 to -0.147 the grade is 6
 from -0.147 to 0.353 the grade is 5
 from 0.353 to 1.059 the grade is 4
 from 1.059 to 1.764 the grade is 3
 from 1.764 to 2.470 the grade is 2
 from 2.470 to 3.176 the grade is 1

If the difference between the colors of the CONFIDENCE INTERVAL COLORS is more than 3 or the msa number (under the column titled MSA) is less than 6, there is insufficient data and an * appears in the COLOR column.

- POS: The position of the acid in the sequence.
- SEQ: The acid one letter.
- ATOM: When there's a model, The ATOM derived sequence in three letter code, including the acid's positions as they appear in the PDB file and the chain identifier.
- SCORE: The normalized conservation scores.
- COLOR: The color scale representing the conservation scores (9 - conserved, 1 - variable).
- CONFIDENCE INTERVAL: When using the bayesian method for calculating rates, a confidence interval is assigned to each of the inferred evolutionary conservation scores, next to it are the colors of the lower and upper bounds of the confidence interval
- B/E: Burried (b) or Exposed (e) residue.
- F/S: functional (f) or structural (s) residue (f - highly conserved and exposed, s - highly conserved and buried).
- MSA DATA: The number of aligned sequences having an acid (non-gapped) from the overall number of sequences at each position.
- RESIDUE VARIETY: The residues variety at each position of the multiple sequence alignment.

POS	SEQ B/E	ATOM F/S	SCORE MSA DATA	COLOR RESIDUE	CONFIDENCE INTERVAL VARIETY
1	R	ARG:1:A	-1.276	9	-1.331, -1.249 9,9
	e	f	89/150	R 98%, K 1%	
2	K	LYS:2:A	-0.239	6	-0.459, -0.071 7,5
	e		102/150	K 42%, R 29%, T 14%, I 2%,	
V 2%, Q	2%, H	1%, M <1%, A <1%, S <1%			
3	P	PRO:3:A	0.195	5	-0.071, 0.373 5,4
	e		117/150	P 52%, K 20%, T 5%, Q 4%,	
F 4%, A	3%, N	2%, I 2%, L 1%, V <1%, R <1%, S <1%, H <1%			
4	I	ILE:4:A	-0.263	6	-0.459, -0.183 7,6
	e		132/150	L 47%, I 27%, F 10%, Y 4%,	
V 3%, M	3%, C	2%, W <1%, T <1%			
5	I	ILE:5:A	-0.554	7	-0.734, -0.459 7,7

	b	135/150	I 49%, V 32%, M 6%, A 6%,
L 3%, C <1%	A	ALA:6:A -0.751	8 -0.897, -0.673 8,7
6	e	f 141/150	A 67%, G 12%, V 9%, I 8%,
M 1%, L <1%	G	GLY:7:A -0.968	8 -1.075, -0.897 9,8
7	b	141/150	G 87%, A 7%, T 4%
8	N	ASN:8:A -1.299	9 -1.331, -1.279 9,9
	e	f 141/150	N 95%, S 4%
9	W	TRP:9:A -1.168	9 -1.249, -1.113 9,9
	b	s 141/150	W 95%, F 3%, M <1%, L <1%
10	K	LYS:10:A -1.322	9 -1.341, -1.306 9,9
	e	f 142/150	K
11	M	MET:11:A -1.262	9 -1.306, -1.249 9,9
	e	f 143/150	M 96%, L 2%, A <1%
12	H	HIS:12:A -0.885	8 -0.990, -0.846 8,8
	e	f 143/150	N 77%, H 13%, Y 4%, F 3%,
S <1%	K	LYS:13:A -0.358	6 -0.536, -0.285 7,6
13	e	143/150	G 45%, K 24%, L 14%, P 5%,
N 3%, M 2%, Y 1%, Q 1%, A <1%, H <1%	T	THR:14:A -0.159	6 -0.376, -0.071 6,5
14	e	143/150	T 39%, S 16%, N 13%, D 9%,
L 7%, M 2%, G 2%, V 2%, I <1%, R <1%, A <1%, E <1%, P <1%, C <1%	L	LEU:15:A 1.115	3 0.579, 1.191 4,3
15	e	143/150	L 24%, R 16%, H 13%, S 6%,
K 5%, P 4%, I 4%, T 4%, A 4%, V 3%, G 3%, Q 2%, N 1%, Y 1%, F 1%	A	ALA:16:A 0.544	4 0.203, 0.579 5,4
16	e	143/150	A 34%, L 11%, S 10%, K 9%,
E 9%, Q 8%, P 2%, D 2%, G 2%, V 2%, T 2%, R 1%, N <1%, I <1%, Y <1%, M <1%	E	GLU:17:A -0.210	6 -0.376, -0.071 6,5
17	e	142/150	E 42%, S 15%, A 11%, Q 9%,
D 7%, M 6%, R 2%, L 1%, H 1%, V <1%, G <1%, I <1%	A	ALA:18:A -0.530	7 -0.673, -0.459 7,7
18	b	142/150	A 44%, L 14%, N 9%, V 7%,
G 6%, T 6%, I 6%, S 2%, E 1%, C <1%, R <1%	V	VAL:19:A 1.065	3 0.579, 1.191 4,3
19	e	142/150	A 20%, V 12%, I 11%, E 9%,
K 8%, R 7%, T 4%, L 4%, S 4%, Q 3%, D 3%, N 2%, M 2%, C <1%, H <1%, G <1%, P <1%	Q	GLN:20:A 0.893	4 0.373, 1.191 4,3
20	e	143/150	E 29%, A 26%, Q 6%, S 6%,
T 5%, R 4%, K 4%, G 4%, D 2%, V 2%, L 2%, H 2%, P <1%, I <1%	F	PHE:21:A -0.120	5 -0.376, 0.056 6,5
21	b	144/150	L 54%, F 20%, A 6%, I 6%,
V 2%, Y 2%, W 2%, H 1%, R <1%, S <1%, M <1%	V	VAL:22:A 0.459	4 0.203, 0.579 5,4
22	b	144/150	V 29%, L 20%, I 14%, A 13%,
C 4%, R 4%, E 3%, D 2%, K 2%, S 1%, G 1%, T 1%, Q <1%, F <1%, M <1%	E	GLU:23:A 0.568	4 0.203, 0.840 5,4
23	e	145/150	A 21%, Q 17%, E 15%, K 11%,

N 7%, D 7%, S 6%, T 4%, G 2%, R 2%, L 1%, H <1%									
24	D	ASP:24:A	1.056	4	0.579,	1.191	4,3		
	e	145/150			G 17%, E 15%, K 13%, A 12%,				
I 10%, D 6%, L 5%, Q 4%, N 4%, S 2%, T 2%, R 2%, V 1%, M 1%, H <1%									
25	V	VAL:25:A	0.056	5	-0.183,	0.203	6,5		
	b	145/150			L 46%, I 17%, V 13%, A 8%,				
F 6%, D 3%, M 1%, T <1%, Y <1%, S <1%, G <1%									
26	K	LYS:26:A	0.712	4	0.373,	0.840	4,4		
	e	145/150			A 26%, K 20%, L 9%, R 8%,				
V 8%, S 6%, N 6%, G 2%, I 2%, T 2%, C 2%, E 2%, D 1%, Q 1%, H <1%									
27	G	GLY:27:A	2.033	2	1.191,	1.732	3,3		
	e	145/150			A 18%, S 11%, D 10%, G 8%,				
K 8%, P 8%, W 7%, E 6%, N 4%, F 4%, Q 2%, H 2%, T 2%, M 1%, Y 1%, L 1%, R <1%									
28	H	HIS:28:A	1.356	3	0.840,	1.732	4,3		
	e	145/150			A 24%, G 15%, L 12%, S 6%,				
Q 6%, H 5%, V 5%, E 5%, T 4%, K 4%, N 2%, M 2%, R 2%, F 1%, I 1%									
29	V	VAL:29:A	1.111	3	0.579,	1.191	4,3		
	b	103/150			L 40%, V 19%, I 10%, A 6%,				
T 2%, M 2%, P 2%, G 2%, H 1%, D 1%, E 1%, Y <1%, Q <1%, F <1%, S <1%, K <1%									
30	P	PRO:30:A	1.821	2	1.191,	1.732	3,3		
	e	136/150			P 16%, A 15%, K 13%, Q 11%,				
D 7%, G 6%, S 6%, R 5%, E 3%, N 2%, T 2%, L 2%, H 2%, V 1%, M <1%, Y <1%, F <1%									
31	P	PRO:31:A	1.650	3	1.191,	1.732	3,3		
	e	145/150			D 21%, L 15%, A 11%, G 8%,				
P 7%, S 7%, E 6%, N 4%, R 4%, I 3%, T 2%, Q 2%, K 2%, H 1%, M <1%, V <1%									
32	A	ALA:32:A	2.967	1	1.732,	3.181	3,1		
	e	146/150			A 15%, D 8%, K 8%, V 8%,				
L 7%, P 7%, G 6%, I 6%, N 6%, E 5%, S 4%, T 4%, R 3%, H 2%, F 1%, Y 1%, Q <1%, M <1%, C <1%									
33	D	ASP:33:A	1.857	2	1.191,	1.732	3,3		
	e	134/150			P 16%, K 11%, D 11%, E 11%,				
S 9%, A 9%, M 5%, L 3%, Q 2%, Y 2%, R 2%, T 2%, V 2%, N 2%, G 2%, H 1%, C <1%, F <1%									
34	E	GLU:34:A	1.343	3	0.840,	1.732	4,3		
	e	146/150			K 17%, G 15%, N 13%, D 10%,				
A 10%, S 7%, E 6%, R 6%, P 3%, Q 2%, T 2%, V 1%, H 1%, L <1%									
35	V	VAL:35:A	0.468	4	0.203,	0.579	5,4		
	b	146/150			V 43%, A 13%, C 10%, T 8%,				
I 8%, R 4%, L 2%, S 2%, N 1%, F 1%, G 1%, Q <1%, K <1%, P <1%									
36	I	ILE:36:A	0.221	5	-0.071,	0.373	5,4		
	e	147/150			E 41%, D 31%, Q 8%, R 3%,				
S 2%, V 2%, I 2%, A 2%, K 1%, H 1%, G 1%, M <1%, Y <1%									
37	S	SER:37:A	-0.166	6	-0.376,	-0.071	6,5		
	b	148/150			V 56%, I 12%, L 6%, A 4%,				
T 4%, S 3%, R 3%, M 2%, P 2%, C 2%, F 1%, K <1%									
38	V	VAL:38:A	-0.010	5	-0.285,	0.056	6,5		
	b	149/150			V 33%, A 26%, L 15%, I 6%,				

F 6%, G 6%, T 2%, M 2%, Y <1%								
39	V	VAL:39:A	-0.169	6	-0.376, -0.071	6,5		
	b	149/150		V 50%, I 30%, L 14%, F 3%,				
A <1%								
40	C	CYS:40:A	-0.145	5	-0.376, -0.071	6,5		
	b	149/150		C 44%, A 22%, I 12%, F 7%,				
L 6%, G 4%, S 1%, V <1%, T <1%								
41	A	ALA:41:A	-0.383	6	-0.607, -0.285	7,6		
	b	149/150		P 70%, V 12%, A 12%, T 2%,				
C 1%, S <1%, M <1%								
42	P	PRO:42:A	-0.648	7	-0.792, -0.536	8,7		
	e	149/150		P 87%, T 4%, S 4%, Q <1%,				
N <1%, V <1%, G <1%, D <1%								
43	F	PHE:43:A	0.226	5	-0.071, 0.373	5,4		
	b	149/150		F 49%, A 26%, Y 8%, S 6%,				
M 2%, P 1%, T 1%, I 1%, V 1%, L <1%, G <1%								
44	L	LEU:44:A	-0.113	5	-0.376, 0.056	6,5		
	e	149/150		T 44%, P 15%, V 15%, L 14%,				
I 8%, M <1%, Q <1%								
45	F	PHE:45:A	-0.104	5	-0.376, 0.056	6,5		
	e	149/150		Y 30%, L 18%, D 15%, A 8%,				
F 7%, S 6%, H 4%, N 4%, C 2%, M <1%, T <1%, G <1%								
46	L	LEU:46:A	-0.535	7	-0.734, -0.459	7,7		
	b	149/150		L 73%, I 18%, V 7%, A <1%				
47	D	ASP:47:A	1.560	3	0.840, 1.732	4,3		
	e	149/150		R 18%, A 15%, S 9%, E 8%,				
P 6%, G 6%, D 5%, H 5%, T 4%, Q 4%, L 4%, I 3%, Y 2%, M 2%, V 1%, F 1%, K <1%, W <1%, N <1%								
48	R	ARG:48:A	0.784	4	0.373, 0.840	4,4		
	e	149/150		S 18%, Q 15%, A 12%, P 8%,				
E 6%, T 6%, R 5%, D 5%, V 4%, L 3%, Y 3%, K 3%, H 2%, G 2%, M 1%, F 1%, I <1%, N <1%								
49	L	LEU:49:A	0.161	5	-0.071, 0.373	5,4		
	b	149/150		V 38%, A 22%, L 10%, M 8%,				
I 4%, T 4%, C 4%, S 2%, F 2%								
50	V	VAL:50:A	0.779	4	0.373, 0.840	4,4		
	e	149/150		A 20%, Q 18%, V 10%, K 10%,				
R 8%, S 6%, L 6%, T 5%, H 2%, I 2%, G 2%, N 2%, M 1%, C <1%, E <1%, D <1%, F <1%, P <1%								
51	Q	GLN:51:A	0.854	4	0.373, 1.191	4,3		
	e	147/150		E 19%, T 14%, A 13%, K 11%,				
Q 7%, S 7%, D 6%, R 6%, N 6%, G 2%, I <1%, H <1%, V <1%, W <1%, P <1%								
52	A	ALA:52:A	1.530	3	0.840, 1.732	4,3		
	e	146/150		L 26%, A 21%, I 6%, T 6%,				
R 6%, V 5%, S 5%, K 5%, E 3%, Q 3%, G 2%, F 2%, N 1%, H 1%, M 1%, P <1%								
53	A	ALA:53:A	0.876	4	0.373, 1.191	4,3		
	b	144/150		L 40%, V 20%, A 10%, I 10%,				
T 6%, C 2%, M 2%, F 2%, S 2%, E 1%, K <1%								
54	D	ASP:54:A	1.339	3	0.840, 1.732	4,3		
	e	129/150		A 20%, K 17%, D 11%, E 10%,				
G 8%, S 8%, R 6%, Q 5%, V 3%, N 2%, H 1%, T 1%, M <1%, P <1%								
55	G	GLY:55:A	0.966	4	0.579, 1.191	4,3		

A 4%, P 56	e	138/150	G 54%, D 10%, S 6%, N 5%,
	T	THR:56:A	-0.126 5 -0.376, -0.071 6,5
E 5%, A 57	e	147/150	S 38%, T 18%, D 14%, K 6%,
	D	ASP:57:A	1.161 3 0.579, 1.191 4,3
G 10%, R 58	e	148/150	K 16%, N 12%, A 11%, P 10%,
	L	LEU:58:A	0.313 5 0.056, 0.373 5,4
A 1%, M 59	b	149/150	I 39%, V 28%, L 24%, F 2%,
	K	LYS:59:A	1.099 3 0.579, 1.191 4,3
T 5%, Q 60	e	149/150	A 25%, K 22%, G 8%, E 5%,
	I	ILE:60:A	0.232 5 -0.071, 0.373 5,4
W 4%, T 61	b	149/150	V 33%, L 26%, I 17%, Y 11%,
	G	GLY:61:A	-0.506 7 -0.673, -0.376 7,6
C 1%, V 62	b	149/150	G 74%, A 16%, S 4%, M 1%,
	A	ALA:62:A	-1.034 9 -1.113, -0.990 9,8
63	s	149/150	A 83%, G 14%, V 1%
	Q	GLN:63:A	-1.186 9 -1.249, -1.150 9,9
64	e	149/150	Q 91%, E 8%
	T	THR:64:A	-1.006 8 -1.113, -0.945 9,8
S <1%, C 65	e	149/150	N 53%, D 40%, T 3%, V <1%,
	M	MET:65:A	-0.113 5 -0.376, 0.056 6,5
I 5%, A 66	b	149/150	C 36%, V 25%, M 18%, L 8%,
	H	HIS:66:A	-0.389 6 -0.536, -0.285 7,6
F 4%, D 67	e	149/150	H 39%, S 28%, Y 12%, A 8%,
	F	PHE:67:A	1.683 3 1.191, 1.732 3,3
Q 6%, E 68	e	149/150	A 15%, P 12%, W 11%, F 10%,
	A	ALA:68:A	0.632 4 0.203, 0.840 5,4
Q 8%, G 69	e	149/150	E 26%, H 17%, A 16%, K 12%,
	D	ASP:69:A	1.022 4 0.579, 1.191 4,3
K 4%, S 70	e	149/150	D 25%, E 22%, A 19%, P 8%,
	Q	GLN:70:A	0.398 4 0.056, 0.579 5,4
Q 6%, N 71	e	149/150	S 42%, A 10%, K 10%, E 8%,
	G	GLY:71:A	-1.304 9 -1.341, -1.279 9,9
72	f	149/150	G
	A	ALA:72:A	-1.270 9 -1.331, -1.249 9,9
73	e	149/150	A 97%, P <1%, S <1%, Q <1%
	Y	TYR:73:A	-0.077 5 -0.285, 0.056 6,5

	b	149/150	Y 39%, F 32%, H 18%, W 4%,
L 2%, R 1%, V <1%, C <1%	T	THR:74:A -1.323	9 -1.341, -1.331 9,9
74	e	f 149/150	T
75	G	GLY:75:A -1.226	9 -1.306, -1.185 9,9
	b	s 149/150	G 98%, S <1%, A <1%
76	E	GLU:76:A -1.093	9 -1.185, -1.034 9,9
	e	f 149/150	E 74%, D 21%, C 1%, Q <1%,
S <1%, N <1%, R <1%	V	VAL:77:A -0.199	6 -0.376, -0.071 6,5
77	b	149/150	I 47%, V 39%, T 7%, L 3%,
N <1%, M <1%, E <1%	S	SER:78:A -1.027	8 -1.113, -0.990 9,8
78	b	149/150	S 85%, A 12%, T <1%, P <1%,
G <1%	P	PRO:79:A -0.329	6 -0.536, -0.183 7,6
79	b	149/150	A 53%, P 24%, G 12%, V 6%,
T 1%, S 1%, I <1%	V	VAL:80:A 1.654	3 1.191, 1.732 3,3
80	e	149/150	A 23%, E 18%, S 12%, P 10%,
V 6%, L 5%, R 4%, T 4%, K 3%, G 3%, D 3%, N 1%, F 1%, Q 1%, I <1%, W <1%, M <1%	M	MET:81:A -1.087	9 -1.185, -1.034 9,9
81	b	s 150/150	M 80%, Q 8%, A 4%, F 4%,
I <1%, L <1%, V <1%	L	LEU:82:A -0.805	8 -0.945, -0.734 8,7
82	b	150/150	L 84%, I 10%, V 3%, A 1%,
W <1%	K	LYS:83:A 0.606	4 0.203, 0.840 5,4
83	e	150/150	K 33%, A 27%, V 8%, R 8%,
Q 6%, T 4%, S 3%, E 2%, L 2%, Y 2%, N 1%, D <1%, G <1%, C <1%, H <1%	D	ASP:84:A -0.399	6 -0.607, -0.285 7,6
84	e	150/150	D 53%, E 22%, K 9%, S 4%,
A 3%, N 2%, T 1%, Q 1%, G 1%, R 1%	L	LEU:85:A 0.764	4 0.373, 0.840 4,4
85	b	150/150	L 28%, A 22%, I 15%, C 10%,
F 8%, V 8%, T 2%, M 2%, Y <1%, S <1%, H <1%, G <1%	G	GLY:86:A -0.222	6 -0.459, -0.071 7,5
86	e	149/150	G 79%, D 6%, N 3%, A 2%,
F 2%, K 1%, L <1%, C <1%, Y <1%, E <1%, S <1%, Q <1%, H <1%, R <1%	V	VAL:87:A -0.240	6 -0.459, -0.071 7,5
87	b	150/150	C 34%, V 28%, A 23%, I 6%,
L 4%, G 2%, T <1%, S <1%	T	THR:88:A 1.327	3 0.840, 1.732 4,3
88	e	150/150	T 19%, R 16%, S 12%, K 12%,
D 12%, E 8%, Q 6%, H 3%, A 3%, P 2%, G 2%, N 2%, C <1%	Y	TYR:89:A 0.401	4 0.056, 0.579 5,4
89	b	150/150	Y 46%, F 15%, W 12%, H 9%,
L 5%, A 4%, I 2%, T 2%, M 1%, G <1%	V	VAL:90:A -0.963	8 -1.075, -0.897 9,8
90	b	150/150	V 86%, A 6%, C 2%, I 2%,
T 1%, S <1%, L <1%	I	ILE:91:A -0.628	7 -0.792, -0.536 8,7
91			

	b	150/150	I 59%, L 17%, V 14%, E 6%,
A 1%, T <1%, M <1%	L	LEU:92:A -0.181	6 -0.376, -0.071 6,5
92	b	150/150	V 38%, L 36%, I 22%, A 2%,
C <1%, T <1%, M <1%	G	GLY:93:A -1.304	9 -1.341, -1.279 9,9
93	b	s 150/150	G
94	H	HIS:94:A -1.321	9 -1.341, -1.306 9,9
	e	f 150/150	H
95	S	SER:95:A -1.315	9 -1.341, -1.306 9,9
	b	s 150/150	S 98%, A 1%
96	E	GLU:96:A -1.324	9 -1.341, -1.331 9,9
	e	f 150/150	E
97	R	ARG:97:A -1.303	9 -1.341, -1.279 9,9
	e	f 150/150	R 99%, C <1%
98	R	ARG:98:A -1.303	9 -1.341, -1.279 9,9
	e	f 150/150	R 99%, S <1%
99	Q	GLN:99:A 0.700	4 0.373, 0.840 4,4
	e	150/150	A 24%, E 20%, Q 18%, T 14%,
H 7%, R 4%, S 3%, L 2%, D 1%, K 1%, V 1%, Y <1%, G <1%, N <1%	M	MET:100:A 0.788	4 0.373, 0.840 4,4
100	e	149/150	Y 31%, D 15%, L 14%, I 12%,
H 6%, M 4%, V 4%, G 2%, F 1%, A 1%, K <1%, Q <1%, S <1%, P	F	PHE:101:A -0.621	7 -0.792, -0.536 8,7
101	e	150/150	F 45%, H 36%, Y 8%, L 2%,
M 2%, I 2%, Q 2%, N <1%, G <1%, R <1%	A	ALA:102:A 2.030	2 1.191, 1.732 3,3
102	e	150/150	G 47%, H 14%, A 12%, N 10%,
R 4%, K 4%, Q 2%, L 1%, C 1%, D <1%, Y <1%, F <1%	E	GLU:103:A -1.260	9 -1.306, -1.249 9,9
103	e	f 150/150	E 97%, D 2%
104	T	THR:104:A -0.287	6 -0.459, -0.183 7,6
	e	150/150	T 54%, S 23%, D 16%, N 1%,
L <1%, R <1%, E <1%, G <1%, K <1%, C <1%	D	ASP:105:A -0.640	7 -0.792, -0.536 8,7
105	e	150/150	D 76%, N 13%, S 6%, E 2%,
A 2%, I <1%	E	GLU:106:A 0.123	5 -0.183, 0.203 6,5
106	e	150/150	E 44%, A 32%, Q 4%, K 3%,
R 3%, D 2%, S 2%, N 2%, V 1%, G 1%, I <1%, L <1%, M <1%, T <1%	T	THR:107:A 1.638	3 1.191, 1.732 3,3
107	e	150/150	L 32%, T 16%, V 12%, I 8%,
F 6%, D 4%, A 4%, Q 4%, M 2%, S 2%, E 1%, R 1%, N 1%, G	V	VAL:108:A -0.476	7 -0.673, -0.376 7,6
<1%, C <1%, W <1%, Y <1%	b	150/150	V 63%, I 24%, L 8%, C 1%,
T 1%, A <1%, E <1%	N	ASN:109:A -0.241	6 -0.459, -0.183 7,6
109	e	150/150	A 34%, N 31%, G 9%, R 9%,
K 3%, L 2%, S 2%, C 2%, Q 1%, T 1%, D <1%, H <1%, I <1%, V	K	LYS:110:A 0.335	5 0.056, 0.579 5,4
<1%, W <1%	e	150/150	A 28%, K 20%, E 13%, L 10%,
110			

R 8%, Q 7%, S 4%, G 2%, D 2%, T 1%, N <1%, V <1%, F <1%, M <1%							
111	K	LYS:111:A	-1.159	9	-1.218,	-1.113	9,9
	e	f	150/150		K 91%, R 6%, Q 2%		
112	V	VAL:112:A	0.189	5	-0.071,	0.373	5,4
	b		150/150		V 33%, A 28%, T 14%, F 7%,		
L 6%, I 6%, Y 1%, G <1%, M <1%							
113	L	LEU:113:A	0.837	4	0.373,	1.191	4,3
	e		150/150		K 25%, A 16%, L 15%, E 13%,		
R 8%, Q 6%, D 3%, S 2%, N 2%, H 2%, M 1%, V 1%, G 1%, P <1%, C <1%							
114	A	ALA:114:A	-0.300	6	-0.459,	-0.183	7,6
	e		150/150		A 61%, K 5%, T 5%, Q 4%,		
H 4%, S 3%, R 3%, V 2%, L 2%, I 2%, Y 2%, M 2%, N 1%, F <1%, C <1%							
115	A	ALA:115:A	-0.822	8	-0.945,	-0.734	8,7
	b		150/150		A 84%, V 6%, I 3%, L 3%,		
C 1%, S 1%, G <1%							
116	F	PHE:116:A	0.393	4	0.056,	0.579	5,4
	b		150/150		L 53%, F 8%, I 8%, H 6%,		
V 6%, Q 5%, R 2%, W 2%, A 2%, K 2%, M 1%, Y <1%, E <1%							
117	T	THR:117:A	1.316	3	0.840,	1.732	4,3
	e		150/150		A 28%, R 16%, D 12%, E 11%,		
K 9%, T 4%, N 4%, S 4%, Q 4%, G 4%, V 1%, H <1%, L <1%							
118	R	ARG:118:A	0.456	4	0.203,	0.579	5,4
	e		150/150		A 37%, H 22%, N 9%, S 5%,		
E 4%, Y 4%, Q 4%, V 3%, R 2%, G 2%, L 1%, I <1%, C <1%, K <1%, T <1%							
119	G	GLY:119:A	0.239	5	-0.071,	0.373	5,4
	e		150/150		G 75%, N 6%, D 6%, E 4%,		
H 1%, Q 1%, K 1%, S 1%, A <1%, P <1%							
120	L	LEU:120:A	-0.403	6	-0.607,	-0.285	7,6
	b		150/150		L 76%, M 14%, I 5%, F 1%,		
A 1%, V <1%							
121	I	ILE:121:A	0.620	4	0.203,	0.840	5,4
	e		150/150		T 37%, I 14%, K 11%, V 11%,		
R 6%, S 4%, G 4%, A 3%, L 2%, E 2%, C 1%, H <1%, N <1%, Q <1%, D <1%, W <1%							
122	P	PRO:122:A	-0.754	8	-0.897,	-0.673	8,7
	b		150/150		P 66%, V 18%, A 11%, I 2%,		
L 1%, S <1%							
123	I	ILE:123:A	-0.865	8	-0.990,	-0.792	8,8
	b		150/150		I 72%, V 16%, L 7%, M 2%,		
T <1%							
124	I	ILE:124:A	0.322	5	0.056,	0.373	5,4
	b		150/150		L 34%, V 32%, I 17%, F 7%,		
A 5%, Y 2%, M 1%							
125	C	CYS:125:A	-1.265	9	-1.331,	-1.249	9,9
	b	s	150/150		C 99%, A <1%		
126	C	CYS:126:A	-0.344	6	-0.536,	-0.285	7,6
	b		150/150		V 53%, I 28%, C 13%, L 3%,		
F 2%							
127	G	GLY:127:A	-1.268	9	-1.331,	-1.249	9,9
	b	s	150/150		G 98%, D 2%		
128	E	GLU:128:A	-1.294	9	-1.331,	-1.279	9,9

129	e	f	150/150	E 98%, D 1%
	S	SER:129:A	-0.204	6 -0.376, -0.071 6,5
	e	148/150	T 58%, S 16%, G 7%, K 4%,	
L 2%, A 2%, E 2%, D 2%, V 1%, Q 1%, P 1%, M <1%				
130	L	LEU:130:A	-0.191	6 -0.376, -0.071 6,5
	e	149/150	L 69%, E 12%, A 6%, Q 4%,	
M 2%, K 1%, I 1%, R 1%, D <1%, G <1%, S <1%				
131	E	GLU:131:A	0.310	5 0.056, 0.373 5,4
	e	149/150	E 44%, A 20%, D 11%, Q 6%,	
S 4%, V 2%, R 2%, T 1%, G 1%, K 1%, N 1%, H <1%, L <1%, P <1%, I <1%, Y <1%				
132	E	GLU:132:A	-0.206	6 -0.376, -0.071 6,5
	e	150/150	E 43%, Q 26%, D 8%, V 8%,	
I 5%, R 2%, T 2%, K <1%, A <1%, S <1%				
133	R	ARG:133:A	-0.894	8 -0.990, -0.846 8,8
	e	f	148/150	R 85%, Y 4%, N 2%, L 2%,
K 2%, Q <1%, T <1%, E <1%, H <1%, G <1%, A <1%				
134	E	GLU:134:A	-0.294	6 -0.459, -0.183 7,6
	e	150/150	E 56%, D 14%, K 10%, Q 8%,	
N 4%, L 2%, V 1%, I 1%, A 1%, R <1%, T <1%				
135	A	ALA:135:A	0.449	4 0.203, 0.579 5,4
	e	150/150	A 49%, S 20%, N 5%, K 5%,	
Q 4%, T 2%, E 2%, L 2%, G 1%, R 1%, D 1%, V 1%, W <1%, Y <1%, P <1%				
136	G	GLY:136:A	-0.013	5 -0.285, 0.203 6,5
	e	150/150	G 75%, N 6%, D 5%, H 4%,	
E 2%, Q 2%, R 1%, K 1%, A <1%				
137	Q	GLN:137:A	1.382	3 0.840, 1.732 4,3
	e	150/150	Q 20%, K 16%, R 14%, E 13%,	
D 6%, T 6%, N 4%, I 4%, L 3%, V 2%, A 2%, S 2%, G 2%, H 1%, M <1%				
138	T	THR:138:A	-0.440	6 -0.607, -0.376 7,6
	e	150/150	T 48%, A 23%, H 9%, Q 6%,	
S 3%, Y 2%, K 1%, P 1%, E <1%, F <1%, G <1%, L <1%, I <1%, M <1%, V <1%				
139	N	ASN:139:A	1.071	3 0.579, 1.191 4,3
	e	150/150	E 22%, V 17%, L 16%, N 8%,	
F 7%, I 6%, K 4%, D 3%, M 2%, A 2%, S 2%, T 2%, Y 1%, Q 1%, G 1%, R <1%				
140	A	ALA:140:A	0.972	4 0.579, 1.191 4,3
	e	150/150	E 26%, A 22%, D 19%, K 8%,	
Q 5%, R 4%, S 3%, N 3%, H 2%, T 2%, I <1%, V <1%, C <1%, M <1%, G <1%, L <1%				
141	V	VAL:141:A	-0.299	6 -0.459, -0.183 7,6
	b	150/150	V 58%, I 12%, H 9%, F 6%,	
Y 4%, W 4%, L 2%, R 1%, E 1%, T <1%				
142	V	VAL:142:A	-0.230	6 -0.459, -0.071 7,5
	b	150/150	V 54%, I 14%, L 12%, C 10%,	
T 4%, P 2%, N 1%				
143	A	ALA:143:A	3.144	1 1.732, 3.181 3,1
	e	150/150	G 12%, L 12%, T 11%, A 10%,	
E 10%, K 10%, S 6%, R 6%, F 5%, V 4%, Q 2%, C 2%, N 2%, D 1%, H <1%, I <1%, M <1%, Y <1%				
144	S	SER:144:A	1.649	3 1.191, 1.732 3,3

e 150/150 A 24%, R 19%, G 13%, S 9%,
 Q 6%, E 5%, N 5%, T 4%, K 4%, D 2%, V 1%, M 1%, H <1%, L
 <1%, P <1%
 145 Q GLN:145:A -1.268 9 -1.306, -1.249 9,9
 e f 150/150 Q 96%, R 2%, S <1%, E <1%
 146 V VAL:146:A -0.044 5 -0.285, 0.056 6,5
 b 150/150 L 52%, I 20%, V 19%, M 4%,
 T 2%, A <1%, F <1%
 147 E GLU:147:A 1.618 3 1.191, 1.732 3,3
 e 150/150 K 16%, E 16%, A 12%, D 11%,
 R 10%, Q 9%, T 5%, S 5%, L 2%, I 2%, N 2%, V 2%, C 2%, G
 <1%, H <1%
 148 K LYS:148:A 0.798 4 0.373, 0.840 4,4
 e 150/150 A 37%, G 22%, E 8%, K 5%,
 S 4%, V 3%, R 3%, D 2%, T 2%, N 2%, I 1%, H 1%, Q 1%, M
 <1%, L <1%, C <1%, P <1%, F <1%
 149 A ALA:149:A -0.258 6 -0.459, -0.183 7,6
 b 150/150 S 26%, A 22%, V 18%, G 15%,
 I 6%, D 4%, L 2%, Q 1%, C 1%, Y <1%, F <1%
 150 L LEU:150:A -0.160 6 -0.376, -0.071 6,5
 b 149/150 L 66%, I 8%, A 8%, V 7%,
 F 5%, T 2%, M 1%, C <1%
 151 A ALA:151:A 1.308 3 0.840, 1.732 4,3
 e 150/150 A 20%, P 19%, D 14%, E 9%,
 K 8%, F 6%, Q 4%, H 2%, N 2%, R 2%, G 2%, S 2%, M 1%, L
 <1%, I <1%, V <1%, Y <1%
 152 G GLY:152:A 1.125 3 0.579, 1.191 4,3
 e 149/150 G 29%, D 14%, A 12%, V 7%,
 K 6%, L 6%, H 5%, E 4%, N 3%, S 2%, R 2%, Q 2%, C 1%, T
 1%, F <1%, P <1%, I <1%
 153 L LEU:153:A 1.032 4 0.579, 1.191 4,3
 b 149/150 L 40%, V 20%, I 11%, G 10%,
 F 3%, T 2%, A 2%, S 2%, D 2%, M 1%, C 1%, N <1%, Q <1%
 154 T THR:154:A 0.940 4 0.579, 1.191 4,3
 e 115/150 S 23%, G 16%, T 15%, D 12%,
 P 6%, E 6%, N 5%, K 3%, R 2%, Q 2%, A 1%, L 1%, C <1%, I
 <1%, H <1%
 155 P PRO:155:A 0.752 4 0.373, 0.840 4,4
 e 111/150 A 37%, E 11%, V 9%, D 6%,
 P 5%, G 5%, K 4%, Q 4%, S 3%, N 3%, I 2%, L 1%, F <1%, R
 <1%, H <1%
 156 E GLU:156:A 1.107 3 0.579, 1.191 4,3
 e 144/150 E 26%, D 20%, A 17%, K 8%,
 S 5%, V 3%, G 3%, T 3%, N 2%, P 2%, Q 2%, F 1%, L 1%, M
 <1%, I <1%
 157 Q GLN:157:A 0.672 4 0.373, 0.840 4,4
 e 139/150 Q 23%, D 19%, A 17%, E 8%,
 T 7%, R 4%, N 3%, S 3%, P 2%, L 1%, G 1%, I 1%, K 1%, C
 <1%, V <1%, M <1%, H <1%
 158 V VAL:158:A 1.395 3 0.840, 1.732 4,3
 b 140/150 A 24%, L 19%, W 12%, V 9%,
 I 7%, F 7%, G 4%, M 3%, P 2%, S 2%, K 2%, R 1%, T 1%, D 1%
 159 K LYS:159:A 0.973 4 0.579, 1.191 4,3
 e 149/150 A 30%, K 12%, E 12%, S 12%,

Q 6%, D 6%, G 4%, T 3%, R 3%, N 2%, P 2%, V 2%, L <1%							
160	Q	GLN:160:A	1.101	3	0.579,	1.191	4,3
	e	149/150					
Q 5%, D 5%, T 4%, S 4%, A 4%, G 3%, P 2%, W 1%, V <1%, H <1%							
161	A	ALA:161:A	0.613	4	0.203,	0.840	5,4
	b	149/150					
A 10%, S 2%, G 2%, C 1%, F 1%, M <1%							
162	V	VAL:162:A	-0.658	7	-0.792,	-0.607	8,7
	b	150/150					
A 2%, S 2%, M 1%, C <1%, R <1%							
163	I	ILE:163:A	-0.263	6	-0.459,	-0.183	7,6
	b	150/150					
M <1%							
164	A	ALA:164:A	-1.252	9	-1.306,	-1.218	9,9
	b	s 150/150					
165	Y	TYR:165:A	-1.261	9	-1.331,	-1.249	9,9
	b	s 150/150					
166	E	GLU:166:A	-1.312	9	-1.341,	-1.306	9,9
	b	s 150/150					
167	P	PRO:167:A	-1.280	9	-1.331,	-1.249	9,9
	b	s 150/150					
168	I	ILE:168:A	-0.921	8	-1.034,	-0.846	9,8
	b	150/150					
L <1%, M <1%							
169	W	TRP:169:A	-1.262	9	-1.331,	-1.249	9,9
	b	s 150/150					
170	A	ALA:170:A	-1.270	9	-1.331,	-1.249	9,9
	b	s 150/150					
171	I	ILE:171:A	-1.276	9	-1.331,	-1.249	9,9
	b	s 150/150					
172	G	GLY:172:A	-1.228	9	-1.306,	-1.185	9,9
	b	s 150/150					
173	T	THR:173:A	-1.090	9	-1.185,	-1.034	9,9
	e	f 150/150					
K <1%, R <1%, E <1%, L <1%							
174	G	GLY:174:A	-1.142	9	-1.249,	-1.075	9,9
	b	s 150/150					
175	K	LYS:175:A	0.026	5	-0.183,	0.203	6,5
	e	150/150					
V 4%, Q 4%, I 2%, M 2%, D 1%, H 1%, T 1%							
176	S	SER:176:A	-0.492	7	-0.673,	-0.376	7,6
	e	150/150					
A 4%, D 1%, I 1%, C <1%, L <1%							
177	S	SER:177:A	-1.095	9	-1.185,	-1.034	9,9
	e	f 150/150					
D <1%, I <1%, V <1%							
178	T	THR:178:A	-0.502	7	-0.673,	-0.376	7,6
	e	150/150					
A <1%, E <1%, L <1%, G <1%, R <1%, N <1%, P <1%, K <1%							
179	P	PRO:179:A	0.746	4	0.373,	0.840	4,4
	e	150/150					
L 3%, V 3%, G 1%, Q 1%, N <1%, C <1%, K <1%, E <1%, I <1%, Y <1%, M <1%, R <1%, W <1%							
180	E	GLU:180:A	0.459	4	0.203,	0.579	5,4

	e	150/150	E 44%, A 17%, D 16%, Q 10%,
S 2%, N 2%, G 2%, K 1%, P 1%, L 1%, F <1%, V <1%			
181	D	ASP:181:A -0.560	7 -0.734, -0.459 7,7
	e	150/150	Q 48%, D 30%, E 5%, Y 4%,
I 3%, M 3%, H 2%, A <1%, T <1%, L <1%, N <1%, W <1%			
182	A	ALA:182:A -0.935	8 -1.034, -0.897 9,8
	b	150/150	A 70%, I 15%, V 11%, S 1%,
L <1%, Q <1%			
183	N	ASN:183:A -0.357	6 -0.536, -0.285 7,6
	e	149/150	Q 46%, E 14%, A 12%, N 11%,
D 5%, V 2%, G 2%, R 2%, L 1%, I 1%, T <1%			
184	S	SER:184:A -0.274	6 -0.459, -0.183 7,6
	e	149/150	E 58%, A 16%, D 10%, S 4%,
K 4%, Q 2%, R 2%, N 1%, T <1%, G <1%			
185	V	VAL:185:A -0.453	7 -0.607, -0.376 7,6
	b	149/150	V 55%, M 20%, A 8%, T 4%,
I 2%, G 2%, R 2%, K 1%, P 1%, Q <1%, L <1%			
186	C	CYS:186:A -0.696	7 -0.846, -0.607 8,7
	b	149/150	H 61%, C 20%, A 5%, I 4%,
Q 2%, L 1%, T 1%, R 1%, D <1%, V <1%, F <1%			
187	G	GLY:187:A 0.819	4 0.373, 0.840 4,4
	e	149/150	A 42%, G 17%, R 8%, K 7%,
S 6%, E 4%, Q 4%, T 2%, V 2%, D 2%, Y 1%, L <1%, I <1%			
188	H	HIS:188:A 1.234	3 0.579, 1.732 4,3
	b	149/150	A 32%, F 25%, L 6%, E 6%,
H 4%, V 4%, Y 3%, Q 2%, S 2%, K 2%, I 2%, G 1%, R 1%, T <1%, D <1%, N <1%, P <1%, C <1%			
189	I	ILE:189:A -0.679	7 -0.792, -0.607 8,7
	b	149/150	I 71%, L 18%, V 5%, M 2%,
A 1%, C <1%			
190	R	ARG:190:A -1.148	9 -1.218, -1.113 9,9
	e	f 149/150	R 93%, K 2%, A 2%, I <1%,
N <1%, T <1%, Q <1%			
191	S	SER:191:A 1.861	2 1.191, 1.732 3,3
	e	149/150	A 24%, E 10%, K 10%, S 8%,
Q 8%, N 8%, G 6%, T 6%, D 6%, L 4%, R 3%, H 1%, P <1%			
192	V	VAL:192:A 1.671	3 1.191, 1.732 3,3
	e	149/150	V 13%, L 11%, Q 11%, R 10%,
T 8%, W 8%, A 7%, E 5%, K 4%, I 3%, H 3%, F 3%, C 2%, Y 2%, S 1%, N <1%, G <1%, M <1%, D <1%			
193	V	VAL:193:A -0.118	5 -0.376, 0.056 6,5
	b	149/150	L 65%, I 15%, V 9%, A 4%,
M 1%, F 1%, C <1%, T <1%, P <1%, G <1%			
194	S	SER:194:A 0.676	4 0.373, 0.840 4,4
	e	148/150	A 37%, V 10%, G 9%, T 9%,
S 8%, K 7%, E 4%, Q 4%, D 2%, R 2%, I 2%, C <1%, L <1%			
195	R	ARG:195:A 1.083	3 0.579, 1.191 4,3
	e	146/150	E 32%, D 12%, K 10%, A 10%,
G 8%, S 6%, R 5%, Q 4%, T 4%, N 2%, V 2%			
196	L	LEU:196:A 1.317	3 0.840, 1.732 4,3
	e	143/150	L 20%, R 19%, K 9%, M 6%,
V 6%, I 6%, Q 5%, N 4%, E 4%, A 4%, T 3%, H 2%, G 1%, F 1%, D <1%, Y <1%, P <1%			
197	F	PHE:197:A 1.219	3 0.579, 1.191 4,3

	e	137/150	Y 25%, F 23%, V 9%, L 8%,
D 8%, S 5%, I 3%, G 3%, H 2%, T 1%, R 1%, N 1%, A 1%, E			
<1%, W <1%, M <1%, P <1%, K <1%			
198	G	GLY:198:A 1.276	3 0.840, 1.732 4,3
	e	131/150	G 42%, S 25%, D 8%, A 5%,
K 3%, T 2%, P 2%, R 2%, E 1%, V 1%, C <1%, H <1%, Y <1%, N			
<1%, Q <1%, I <1%			
199	P	PRO:199:A 2.765	1 1.191, 3.181 3,1
	e	132/150	E 16%, A 15%, D 15%, P 13%,
K 7%, Q 7%, S 6%, T 6%, G 3%, R 2%, F 1%, N <1%, H <1%, V			
<1%, I <1%, W <1%, L <1%			
200	E	GLU:200:A 0.833	4 0.373, 1.191 4,3
	e	129/150	E 31%, A 19%, D 13%, G 7%,
V 7%, T 4%, S 3%, I 3%, Q 3%, K 2%, N 1%, L <1%			
201	A	ALA:201:A 1.437	3 0.840, 1.732 4,3
	b	82/150	V 26%, A 19%, T 15%, I 9%,
L 9%, E 3%, H 2%, M 2%, S 1%, C 1%, G 1%, F 1%, R 1%, Q			
1%, W 1%, D 1%			
202	A	ALA:202:A -0.613	7 -0.792, -0.536 8,7
	e	147/150	A 64%, G 23%, S 7%, K 1%,
L 1%, P <1%, W <1%, T <1%			
203	E	GLU:203:A 1.307	3 0.840, 1.732 4,3
	e	149/150	D 20%, E 20%, A 18%, S 7%,
Q 6%, N 6%, K 4%, G 4%, R 4%, P 3%, H 2%, T 2%			
204	A	ALA:204:A 3.176	1 1.732, 3.181 3,1
	e	149/150	A 14%, S 11%, G 11%, K 10%,
D 8%, N 8%, R 8%, E 7%, Q 6%, T 6%, V 3%, H 2%, L 1%, M			
<1%, P <1%, I <1%			
205	I	ILE:205:A 0.244	5 -0.071, 0.373 5,4
	b	149/150	V 42%, I 18%, T 15%, L 9%,
M 8%, A 2%, R 1%, P 1%, C <1%			
206	R	ARG:206:A -0.479	7 -0.673, -0.376 7,6
	b	149/150	R 65%, S 7%, P 7%, I 4%,
T 3%, A 3%, Q 2%, H 1%, K 1%, G 1%, L <1%, Y <1%			
207	I	ILE:207:A -0.759	8 -0.897, -0.673 8,7
	b	148/150	I 73%, V 12%, L 12%, C 2%
208	Q	GLN:208:A -0.877	8 -0.990, -0.792 8,8
	b	148/150	L 69%, Q 20%, I 6%, V 2%,
M 1%			
209	Y	TYR:209:A -1.314	9 -1.341, -1.306 9,9
	s	148/150	Y
210	G	GLY:210:A -1.266	9 -1.331, -1.249 9,9
	s	148/150	G 99%, A <1%
211	G	GLY:211:A -1.304	9 -1.341, -1.279 9,9
	s	148/150	G
212	S	SER:212:A -1.315	9 -1.341, -1.306 9,9
	f	148/150	S 99%, P <1%
213	V	VAL:213:A -1.008	8 -1.113, -0.945 9,8
	e	148/150	V 84%, M 8%, A 3%, C 1%,
L 1%, I <1%			
214	K	LYS:214:A -0.688	7 -0.846, -0.607 8,7
	e	148/150	K 63%, N 18%, T 10%, S 4%,
G 2%			
215	P	PRO:215:A 0.487	4 0.203, 0.579 5,4

	e	148/150	P 50%, A 22%, S 8%, G 7%,
D 2%, E 2%, I 1%, T 1%, K 1%, L 1%, V <1%, F <1%			
216	D	ASP:216:A 1.427	3 0.840, 1.732 4,3
	e	149/150	D 20%, G 18%, A 16%, S 13%,
E 12%, K 6%, N 4%, Q 4%, T 2%, Y <1%, R <1%, H <1%			
217	N	ASN:217:A -1.133	9 -1.218, -1.113 9,9
	e	f 149/150	N 89%, T 6%, S 1%, D 1%,
L <1%			
218	I	ILE:218:A -0.622	7 -0.792, -0.536 8,7
	b	148/150	A 59%, I 15%, V 10%, C 7%,
S 3%, T 1%, N <1%, L <1%, F <1%, X <1%			
219	R	ARG:219:A 0.307	5 0.056, 0.373 5,4
	e	149/150	A 40%, K 20%, R 6%, E 6%,
V 6%, S 4%, G 3%, D 3%, T 2%, Q 1%, L 1%, N <1%, Y <1%, M <1%, P <1%, H <1%			
220	D	ASP:220:A -0.334	6 -0.536, -0.285 7,6
	e	149/150	E 62%, D 10%, A 7%, G 4%,
Q 4%, S 2%, K 1%, R 1%, L 1%, P 1%, V <1%, W <1%, T <1%, M <1%			
221	F	PHE:221:A -0.458	7 -0.607, -0.376 7,6
	b	149/150	L 57%, I 32%, F 6%, Y 2%,
Q <1%, V <1%			
222	L	LEU:222:A -0.151	6 -0.376, -0.071 6,5
	b	147/150	L 27%, F 26%, M 23%, A 8%,
I 5%, V 3%, G 1%, S 1%, C <1%, T <1%			
223	A	ALA:223:A 0.351	5 0.056, 0.579 5,4
	e	147/150	A 45%, S 12%, G 10%, K 7%,
Q 4%, E 4%, N 3%, T 3%, R 2%, D 2%, C 1%, H 1%, P <1%, I <1%			
224	Q	GLN:224:A 0.233	5 -0.071, 0.373 5,4
	e	149/150	Q 34%, M 14%, K 12%, V 10%,
L 8%, E 6%, T 2%, C 2%, I 2%, G 2%, A 1%, S 1%, N 1%			
225	Q	GLN:225:A 1.514	3 0.840, 1.732 4,3
	e	148/150	P 48%, A 10%, D 10%, E 8%,
S 5%, Q 4%, K 2%, R 2%, T 2%, V 2%, N 1%, G 1%, L <1%			
226	Q	GLN:226:A -0.343	6 -0.536, -0.285 7,6
	e	148/150	D 62%, N 14%, E 10%, H 4%,
G 4%, S 1%, Q <1%, R <1%, P <1%, A <1%			
227	I	ILE:227:A -0.852	8 -0.945, -0.792 8,8
	b	148/150	V 50%, I 47%, C <1%, N <1%
228	D	ASP:228:A -1.088	9 -1.185, -1.034 9,9
	e	f 148/150	D 88%, N 5%, G 5%, S <1%
229	G	GLY:229:A -1.304	9 -1.341, -1.279 9,9
	b	s 148/150	G
230	P	PRO:230:A -0.420	6 -0.607, -0.285 7,6
	b	148/150	A 45%, G 27%, L 10%, F 10%,
V 5%, C 1%, P <1%			
231	L	LEU:231:A -1.290	9 -1.331, -1.279 9,9
	b	s 148/150	L 93%, F 6%
232	V	VAL:232:A -0.877	8 -0.990, -0.792 8,8
	b	148/150	V 70%, I 27%, L 2%
233	G	GLY:233:A -1.304	9 -1.341, -1.279 9,9
	b	s 148/150	G
234	G	GLY:234:A -0.963	8 -1.075, -0.897 9,8
	b	147/150	G 87%, R 8%, A 1%, S 1%,
D <1%, N <1%			

235	A	ALA:235:A	-1.188	9	-1.249, -1.150	9,9
	b	s	146/150	A 91%, S 4%, C 1%, H <1%,		
D <1%, Q <1%, T <1%						
236	S	SER:236:A	-1.204	9	-1.279, -1.185	9,9
	b	s	146/150	S 89%, A 10%, T <1%		
237	L	LEU:237:A	-1.169	9	-1.249, -1.113	9,9
	b	s	146/150	L 91%, W 5%, I 2%		
238	E	GLU:238:A	0.410	4	0.056, 0.579	5,4
	e		146/150	K 39%, D 21%, N 9%, E 7%,		
V 6%, Q 6%, T 4%, S 2%, L 1%, R 1%						
239	P	PRO:239:A	-0.203	6	-0.376, -0.071	6,5
	b		138/150	A 62%, P 17%, V 5%, G 3%,		
S 2%, L 2%, E 1%, K <1%, F <1%, T <1%, C <1%, I <1%, N <1%						
240	A	ALA:240:A	1.008	4	0.579, 1.191	4,3
	e		142/150	E 26%, D 23%, A 16%, P 7%,		
K 5%, Q 4%, N 4%, V 3%, G 2%, S 2%, R 1%, M <1%, L <1%, T <1%						
241	S	SER:241:A	-0.096	5	-0.285, 0.056	6,5
	e		144/150	D 34%, E 25%, S 19%, G 6%,		
Q 3%, K 3%, A 3%, T 1%, H <1%, V <1%, P <1%, N <1%						
242	F	PHE:242:A	-1.167	9	-1.249, -1.113	9,9
	b	s	144/150	F 93%, Y 5%, L <1%		
243	L	LEU:243:A	0.698	4	0.373, 0.840	4,4
	b		143/150	L 33%, A 18%, V 16%, I 7%,		
S 6%, G 4%, T 2%, F 2%, M 2%, N 1%, W 1%, K <1%, E <1%, Y <1%, C <1%						
244	Q	GLN:244:A	0.660	4	0.373, 0.840	4,4
	e		143/150	A 32%, K 11%, D 9%, Q 7%,		
E 6%, G 6%, S 5%, P 5%, T 4%, R 4%, N 2%, I <1%, V <1%						
245	L	LEU:245:A	-1.078	9	-1.150, -1.034	9,9
	b	s	139/150	I 84%, L 13%, M 1%, V <1%		
246	V	VAL:246:A	0.331	5	0.056, 0.579	5,4
	b		120/150	V 35%, I 28%, C 17%, A 9%,		
L 7%, T <1%, E <1%, Y <1%						
247	E	GLU:247:A	0.575	4	0.203, 0.840	5,4
	e		92/150	R 17%, E 16%, N 15%, Q 14%,		
K 11%, A 10%, D 4%, S 3%, V 3%, T 1%, I 1%, G 1%						
248	A	ALA:248:A	-0.852	8	-0.990, -0.734	8,7
	e	f	68/150	A 82%, Y 4%, F 4%, L 2%,		
S 2%, G 1%, V 1%						
249	G	GLY:249:A	0.043	5	-0.376, 0.203	6,5
	e		32/150	A 46%, G 25%, Q 9%, K 6%,		
W 3%, E 3%, T 3%, Y 3%						
250	R	ARG:250:A	0.953	4	0.203, 1.191	5,3
	e		16/150	R 31%, K 31%, A 25%, N 6%,		
I 6%						
251	H	HIS:251:A	-0.614	7	-0.990, -0.376	8,6
	e		10/150	H 80%, N 20%		
252	E	-	-0.806	8*	-1.249, -0.607	9,7
	e	f	3/150	E		

*Below the confidence cut-off – The calculations for this site were performed on less than 6 non-gapped homologue sequences, or the confidence interval for the estimated score is equal to– or

larger than- 4 color grades.